Superstore Sales: Forecast, Risk & Segmentation Briefing

Prepared for the Head of Supply Chain and the Chief Financial Officer • Prepared by Kuchuru Sai Krishna Reddy, XYLOFY AI Internship Program • July 2026

Executive Summary

Over the last four years, the business generated $2.26 million in sales across 4,922 orders, and the trend line points steadily upward, so growth is real and continuing. We built and compared three different forecasting methods and found one, a technique called XGBoost, consistently predicts monthly sales more accurately than the others, with a typical error of about 15%. Using that method, we expect combined sales to run roughly $50,000 in January, $52,000 in February, and climb to about $74,000 in March, driven mainly by Office Supplies and Technology. We also flagged four unusual months in the sales history worth investigating, identified four distinct product groups that each need a different stocking approach, and found that the West region grows the most reliably while the East region swings the most. None of this requires technical background to act on: the short version is grow inventory into March, double-check the biggest sales spikes on record, and treat the West region as the template for stable planning.

Key Findings from Data Exploration and Forecasting

- **Category mix is balanced.** Technology is the top revenue category at $827,456 (36.6% of sales), followed by Furniture at $728,659 (32.2%) and Office Supplies at $705,422 (31.2%) — the three categories are close in size, so no single one dominates the business.
- **Regional performance is uneven.** The West region produced the most total revenue ($710,220) with the steadiest month-to-month growth, while the East region posted the single highest peaks but the largest swings between good and bad months. Central and South trail both at $492,647 and $389,151.
- **Sales are seasonal.** March is consistently the strongest sales month of the year, with October, November, and December also running well above average; February is consistently the weakest month across all four years.
- **Shipping is consistent.** Average delivery time is about four days and is nearly identical across all four regions, so shipping speed is not currently a source of regional sales differences.
- **XGBoost is the most accurate forecasting method.** We tested three forecasting techniques (SARIMA, Prophet, and XGBoost). XGBoost was the clear winner, with a typical monthly forecasting error of 14.8%, versus roughly 22% for the other two methods, so it is the model we recommend for ongoing use.

3-Month Sales Forecast (in Plain Language)

The table below shows what we expect total sales to be for each of the next three months, along with a realistic low-to-high range that accounts for the model's typical margin of error (about 15%). Think of the range as "we'd be surprised if the real number fell outside this band" rather than an exact prediction.

Month	Expected Sales	Likely Range (Low – High)
January	$49,600	$42,300 – $56,900
February	$52,500	$44,700 – $60,300
March	$74,000	$63,000 – $85,000

March's jump is driven by Office Supplies (forecast to reach about $29,500 that month) and Technology (about $27,600), both of which are trending upward. Furniture is forecast to stay the smallest contributor of the three, at roughly $16,900 in March.

Top 3 Sales Anomalies and Likely Causes

- **November 2018 (unusually HIGH).** Sales hit $117,938 that month, nearly two-and-a-half times the average — the single biggest month on record. Likely cause: Black Friday and Cyber Monday promotions combined with year-end holiday buying.
- **December 2017 (unusually HIGH).** Sales reached $95,739, well above the typical month. Likely cause: year-end holiday demand and corporate customers spending remaining annual budgets before the calendar year closed.

- **February 2015 (unusually LOW).** Sales dropped to just $4,520, the lowest point in the entire four-year history. Likely cause: a seasonal post-holiday slowdown, compounded by this being early in the business's operating history when order volume was still low.

A fourth, smaller dip also occurred in February 2016 ($11,951), reinforcing that February is a structurally slow month rather than a one-off event. We recommend confirming the November and December spikes against known promotional calendars, since correctly separating "expected holiday surge" from "data or fulfillment issue" changes how much safety stock is needed going forward.

Product Demand Segmentation and Stocking Strategy

Segment	Products	Recommended Stocking Strategy
High-Revenue Premium	Copiers, Machines	Keep dedicated safety stock; each sale is high-value, so a stockout is costly even though these sell less often.
Core Business	Chairs, Phones, Storage, Tables	Highest overall sales — prioritize availability and the tightest, most frequent replenishment cycle.
Fast-Growing Everyday	Accessories, Appliances, Art, Binders, Labels, Paper	Demand is rising fastest here — increase stock levels ahead of season and monitor closely to avoid running out.
Low-Demand	Bookcases, Envelopes, Fasteners, Furnishings, Supplies	Keep lean, just-in-time stock; tying up capital here has the lowest payoff.

Business Recommendations

- **1. Build inventory ahead of the Office Supplies and Technology climb into March.** Office Supplies is forecast to grow from about $24,700 to $29,500 a month (+19%) and Technology from about $17,600 to $27,600 (+57%) between January and March. Increase purchase orders and warehouse space for these two categories now so stock is in place before the March peak.
- **2. Plan a deliberate Q4 surge strategy instead of standard stocking.** The two largest sales spikes on record both landed in November and December, and the broader seasonal pattern confirms October–December run well above average every year. Set safety-stock and staffing levels for the Core Business and High-Revenue Premium segments (Chairs, Phones, Storage, Tables, Copiers, Machines) higher for Q4 specifically, rather than using a flat year-round buffer.
- **3. Treat West as the planning benchmark and investigate East's volatility before expanding there.** West delivered the highest total revenue ($710,220) with the steadiest growth, while East had the highest peaks but the widest swings. Use West's pattern as the reliable baseline for company-wide planning, and run a short root-cause review of what drives East's volatility (promotions, stock availability, or demand) before increasing inventory investment there to match West.

Risk and Limitation to Keep in Mind

The forecast is built entirely from the company's own four years of order history (2015–2018) plus the calendar — it does not know about planned promotions, price changes, competitor moves, supplier disruptions, or broader economic shifts, because none of that is in the data it learned from. Combined with a typical error margin of about 15%, this means the forecast should be treated as a directional planning tool, not a guaranteed number, especially for any month that includes an event the business hasn't seen the like of before. We recommend refreshing the model with new sales data every quarter and reviewing forecasts against real bookings before committing to large inventory purchases.